

**TEMPLE UNIVERSITY
FOX SCHOOL OF BUSINESS**

**AS 3502 / AS 5103 Advanced Long-Term Actuarial Modeling
Fall 2026**

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The course introduces multi-state models to extend the pricing and reserving concepts of long-term actuarial modeling to different states of the environment, multiple decrements (such as modeling different health statuses), and multiple lives (such as life insurance and annuity products for a couple or business partners). The course also includes actuarial applications to profit testing and pension valuation. Fund values and profit testing for universal life products and equity-linked insurance are discussed as well as embedded options in life insurance and annuity products.

CLASS MEETING TIMES

Mondays, Wednesdays 2:00 p.m. – 3:15 p.m.

Alter 237

COURSE & EXAM MATERIALS

- Textbook: Actuarial Mathematics for Life Contingent Risks, 3rd edition, by Dickson, Hardy, and Waters. ISBN: 978-1-108-47808-3
- Problem Sets (Viswanathan)
- SOA Sample Questions

STUDENT DROP-IN HOURS

Mondays 3:30 p.m. – 4:30 p.m. (629 Alter)

Wednesdays 11:00 a.m. – 12:00 p.m. (629 Alter)

Thursdays 10:00 a.m. – 12:00 p.m. (Zoom)

I am available at other times during the week so please contact me to schedule an appointment for another time. These appointments can be over Zoom or in-person.

LEARNING GOALS AND ASSURANCE OF LEARNING

The four main learning goals for the Bachelor of Business Administration Program are:

- **Demonstrate business knowledge needed to make business decisions.**
 - Apply a core body of discipline-specific knowledge to business situations and problems.
- **Apply critical thinking skills to business decisions.**
 - Use integrated business knowledge to identify problems, generate solutions, and develop reasoned conclusions.
 - Understand the ethical, legal, and social responsibilities of individuals and organizations.
- **Apply quantitative reasoning skills to make recommendations and business decisions.**
 - Apply quantitative analysis and interpretation to business problems.
 - Use data to analyze business decisions.
 - Use software to analyze and implement business decisions.
- **Communicate Professionally.**
 - Demonstrate professional business writing skills.
 - Demonstrate professional business presentation skills.

COURSE GUIDELINES

- Kyra Nguyen (kyra.nguyen@temple.edu) is the Peer Teacher for this course. Please attend her drop-in hours for help with this material, with homework assignments, and in preparing for exams.
- The material in this course is tested on SOA Exam ALTAM Advanced Long-Term Actuarial Mathematics and there is an opportunity to earn UEC Exam ALTAM credit through this course.
- You will have free access to the classroom version of Coaching Actuaries for Exam ALTAM.
- If you are interested in purchasing additional exam materials from Coaching Actuaries such as ADAPT to prepare for Exam ALTAM, there are discounts for Temple students.

CLASSROOM GUIDELINES

- Please bring an SOA approved calculator to each class session. You may only use SOA approved calculators for the exams in this course. I recommend the TI-30XS Multiview calculator.
- There will be a problem file posted before most class sessions containing problems to be discussed that day.
- You are expected to attend each class and take notes. This will help you in mastering this challenging subject material. You should also visit Canvas view the notes and any assignment.
- To facilitate your understanding of this material, reading through your notes after class and prior to the next meeting of this course is highly encouraged. You should

also read the assigned textbook sections. Most importantly, please email me or see me if you need help understanding or applying any of the concepts.

- Additional practice questions will be posted on Canvas on many Monday and Wednesday afternoons, with the solution files posted the day after. It is important for your progress to attempt these questions, thereby strengthening your problem-solving skills. You can email the professor for any help on these problems or to check your solutions.
- Problem sets, homework assignments, and grades will be posted on Canvas.

HOMEWORK GUIDELINES

- Homework assignments must be submitted through Canvas by 2:00 p.m. on the due date. Assignments must show all work with the **final answer circled or indicated**.
- Homework must be uploaded as a pdf (multi-page pdf if applicable). You can use a tablet to solve the homework or write your work on paper. Please don't solve problems on the homework assignment (by hand or on a tablet) as there isn't sufficient room to fully show all work. If writing on paper, you can then scan your work into a pdf. I highly recommend obtaining the Adobe Scan app. Given Temple's Adobe Creative Cloud license, you have access to this and other Adobe products through both app and desktop platforms. Make sure to write your name on your homework.
- If your assignment is not submitted 2:00 p.m. on the due date, it is considered late. You have 24 hours after 2:00 p.m. on the due date to submit the late assignment; late homework assignments will receive a grade penalty of 25%.
- After you submit your file through Canvas, check the uploaded file to see that it contains all your work and is legible.
- You are expected to work on homework questions yourself, however you may seek assistance from the professor, the Peer Teacher, or fellow classmates.
- Your raw score for the homework portion of your final grade is calculated as the sum of points you received on the homework assignments divided by the sum of the total possible points on the homework assignments.

EXAMS

The exams in this course will be given in class. The tentative dates for Exam #1 and Exam #2 are listed in the schedule on the last pages of this syllabus. Those dates will be finalized in class. The final exam in the course will be given on:

Wednesday, December 14, 2026 1:00 p.m. – 3:00 p.m.

All exams are closed book, closed notes. During our exams, you may not work with anyone or obtain help from anyone. *Actuaries are known for their ethics and honesty. Preserve your reputation through your ethical behavior on exams and assignments in our course.*

COURSE GRADE AND UEC

The final numerical grade is a weighted average of these components.

	Weights for Course Grade	Weights for UEC
Homework	20%	15%
Project	10%	5%
Exam #1	20%	15%
Exam #2	20%	15%
Final Exam	30%	50%

The Final Exam in this course is cumulative.

The final numerical grade based on the UEC weights will be used to determine whether the student qualifies for Exam ALTAM credit. A student must receive a UEC grade of 85% to qualify for UEC ALTAM credit.

EXAM MAKEUP POLICY

1. If a student misses an exam, for that exam to count for UEC grading, they must take the approved exam within seven days of the original exam date. Official documentation for missing the exam must be provided.
2. After seven days, students may still take the approved exam for the course grade but only in very rare cases will that exam count for UEC grading. This will be subject to the SOA External Examiner's approval.
3. If the exam is taken substantially later than the original exam date, students could shift the exam weight to the final exam weight. According to UEC rules, students will lose potential UEC credit eligibility.

TEMPLE COURSE GRADE

Final Numerical Grade	Corresponding Temple Letter Grade
93.5% - 100%	A
90.0% - 93.4%	A-
87.0% - 89.9%	B+
83.5% - 86.9%	B
80.0% - 83.4%	B-
77.0% - 79.9%	C+
73.5% - 76.9%	C
70.0% - 73.4%	C-
67.0% - 69.9%	D+
63.5% - 66.9%	D
60.0% - 63.4%	D-
<60.0%	F

UNIVERSITY POLICIES

EXPECTATIONS FOR CLASS CONDUCT

It is important to foster a respectful and productive learning environment that includes all students in our diverse community of learners. Our differences, some of which are outlined in the University's nondiscrimination statement, will add richness to this learning experience. Therefore, all opinions and experiences, no matter how different or controversial they may be perceived, must be respected in the tolerant spirit of academic discourse.

Treat your classmates and instructor with respect in all communication, class activities, and meetings. You are encouraged to comment, question, or critique an idea but you are not to attack an individual. Please consider that sarcasm, humor and slang can be misconstrued in online interactions and generate unintended disruptions. Profanity should be avoided as should the use of all capital letters when composing responses in discussion threads, which can be construed as "shouting" online. Remember to be careful with your own and others' privacy. In general, have your behavior mirror how you would like to be treated by others.

ACCOMMODATIONS

Any student who has a need for accommodations based on the impact of a documented disability or medical condition should contact Disability Resources and Services (DRS) located in the Howard Gittis Student Center South, 4th Floor at drs@temple.edu or 215-204-1280 to request accommodations and learn more about the resources available to you. If you have a DRS accommodation letter to share with me, or you would like to discuss your accommodations, please contact me as soon as practical. I will work with you and with DRS to coordinate reasonable accommodations for all students with documented disabilities. All discussions related to your accommodations will be confidential. Students can learn more about the accommodation process and pre-register on the [DRS website](#). Students may register at any time during the semester, but accommodations are not active until you register, so I recommend doing so as early in the semester as possible.

ATTENDANCE

To achieve course learning goals, students must attend and participate in classes, according to the course requirements. However, if you have tested positive for or are experiencing symptoms of a contagious illness, you should not come to campus or attend in-person classes or activities. Please contact me to create a plan for participation and engagement in the course as soon as you are able to do so, and to make a plan to complete all assignments in a timely fashion.

SCHOOL-LIFE BALANCE

Many unpredictable things can happen over the course of a semester. If you find yourself struggling to balance your education and your other commitments, please reach out to me immediately so that we can work together to build a plan for your success. Some of the resources in the next section may also be of use to you.

RESOURCES AND SUPPORT

Sometimes the biggest factors impacting student success are things happening beyond the scope of the individual classroom. Temple provides a wide array of resources both to help you overcome academic challenges and those not directly related to the educational challenges of the course. Please reach out to me if you need help deciding which resources might be right for you.

- [Student Success Center](#)
- [University Libraries](#)
- [Undergraduate Research Support](#)
- [Career Center](#)
- [Tuttleman Counseling Services](#)
- [Disability Resources and Services](#)
- [Student Health Services](#)

If you are experiencing food insecurity or financial struggles, Temple provides resources and support. Notably, the Temple University [Cherry Pantry](#) is in operation as well as a variety of resources from the [Division of Student Affairs](#).

TECHNOLOGY GUIDELINES

To participate in this course, you will need the equipment, software, and internet access necessary to reliably use Canvas as well as other applications. Limited resources are available for students who do not have the technology they need for class. Students with educational technology needs, including no computer or camera or insufficient Wifi access, should submit a Student Technology Assistance Application located in TUPortal and linked from the [Dean of Students Support and Resources](#) webpage. The university will endeavor to meet needs, such as with a long-term loan of a laptop or Mifi device, a refurbished computer, or subsidized internet access.

Note that there are technology resources available for students, including on-campus computers available for student use, [the Tech Center](#) computer labs and free [laptop and battery borrowing](#) at Charles Library, software that is available for free download and other specialty software that may be available for remote access through ITS.

RECORDINGS

Any recordings permitted in this class can only be used for the student's personal educational use. Students are not permitted to copy, publish, or redistribute audio or video recordings of any portion of the class session to individuals who are not students in the course or academic program without the express permission of the faculty member and of any students who are recorded. Distribution without permission may be a violation of educational privacy law known as FERPA as well as certain copyright laws. Any recordings made by the instructor or university of this course are the property of Temple University.

ACADEMIC GUIDELINES

ACADEMIC FREEDOM

Freedom to teach and freedom to learn are inseparable facets of academic freedom. I have the freedom and responsibility to design and facilitate our learning environment to best achieve the promise of the course as outlined in its official description. You have the responsibility to engage with the course in good faith and freedom from mistreatment when your opinion differs from mine. Note that it is not abuse of this freedom for me to require that you support relevant opinions with clear argumentation and solid evidence. For more on academic freedom, consult the [official Temple policy](#) on the matter.

ACADEMIC INTEGRITY

Temple University believes strongly in academic honesty and integrity. Plagiarism and academic cheating are, therefore, prohibited. All work you submit for assessment should be your own efforts. For more on this topic, consult the relevant portions of [Temple Bulletin](#) and the [Student Conduct Code](#).

SOA ALTAM Syllabus

- *Actuarial Mathematics for Life Contingent Risks*, Third Edition Dickson, C.M.D., Hardy, M.R., Waters, H.R. (2020), Cambridge University Press ISBN: 978-1-108-47808-3. Exercises are considered part of the required readings.
 - Chapter 3 (Sections 4 and 10)
 - Chapter 7 (Sections 2.4 and 4)
 - Chapter 8
 - Chapter 9
 - Chapter 10
 - Chapter 11 (except section 12)
 - Chapter 13 (except section 8)
 - Chapter 14
 - Chapter 15 (sections 1-3)
 - Chapter 17
 - Chapter 18 (Sections 5 and 6) [Note: sections 1 through 3 are background only]
- [ALTAM-21-23: Variable Annuity Guarantees](#)

SCHEDULE

	Date	Textbook	Topic
M	8/24/2026	8.1 – 8.3	Multi-State Models, Discrete Markov Chains
W	8/26/2026		Continuous Markov Chains, Markov Chain applications
M	8/31/2026	8.4, 8.4.1	Kolmogorov Forward Equations, Euler's Method
W	9/2/2026		Actuarial Present Value, Premiums, Policy Values
M	9/7/2026	8.6, 8.7, 8.9	Labor Day (no classes held)
W	9/9/2026		Reserves, Long-Term Care Insurance, Continuing Care Retirement Communities
M	9/14/2026	9.1 – 9.4	Multiple Decrement Models: Probability Models
W	9/16/2026		Multiple Decrement Models: Discrete Probability, Life Tables
M	9/21/2026	9.5 – 9.7	Multiple Decrement Models: Associated Single Decrement
W	9/23/2026	7.4, 8.8	Multiple Decrement Models: UDD and CFM
M	9/28/2026		Multiple Decrement Models: Exact Age transitions
W	9/30/2026		Thiele's equation for single life and multi-state models
M	10/5/2026		Exam #1*
W	10/7/2026	3.4, 3.10, 10.1 – 10.4	Heterogeneity and Trends in Mortality, Multiple Lives: Joint Life & Last Survivor
M	10/12/2026		Multiple Lives: Contingent Functions and Common Stock
W	10/14/2026	10.5 – 10.7	Multiple Lives: Insurance and Annuities
M	10/19/2026		Common Shock Insurance and Annuities, Reversionary Annuities
W	10/21/2026	13.1 – 13.9	Profit Analysis: Cash Flows, Profit Vector, Signature, Profit Measure
M	10/26/2026		Profit Analysis: Premiums, Reserves, Multi-State Models, Gain by Source, Salary Projections
W	10/28/2026	11.1 – 11.7	Pension Plans: Pension Overview, Salary Projections, DC Plans, DB Plans Valuation of Benefits
M	11/2/2026	11.8 – 11.11	Exam #2*
W	11/4/2026	14.1 – 14.3	Pension Plans: TUC and PUC
M	11/9/2026		Thiele's, Universal Life: Type A and Type B
W	11/11/2026		Universal Life: Cashflows
M	11/16/2026	17.1 – 17.5	Universal Life: Policy Values
W	11/18/2026		Guaranteed Min Death Benefit Payoffs, Black-Scholes
M	11/30/2026	17.6 – 17.7, 15.1 – 15.3	Guaranteed Min Accumulation/Maturity Benefit Payoffs, Black-Scholes
W	12/2/2026		Guaranteed Min Income, Withdrawal Benefit
M	12/7/2026		Delta-Hedging

* Exam dates are tentative and will be finalized in class.